

计算机类培养方案（2025 级）

一、大类所包含的专业

计算机类包含如下专业：计算机科学与技术；软件工程；智能科学与技术

二、大类阶段课程设置

大类学科基础课程（必修）：高等数学 A(一)、计算机科学导论、C 语言程序设计

大类学科选修课程（选修）：无

大类实践课程：C 语言程序设计实验、军事训练

软件工程专业培养方案

Curriculum for Undergraduate of Software Engineering Major

一、专业介绍（200 字左右）

本专业从 2005 年开始招收本科生，属于工学一级学科，拥有软件工程一级学科硕士学位授予权，湖北省“十二五”重点特色学科和省级战略性新兴产业（支柱）产业人才培养计划专业，湖北省服务外包人才培养（训）基地，湖北省“荆楚卓越人才”协同育人计划专业，校级“3+1+2”拔尖人才培养计划专业，首批国家级一流本科专业建设点。专业以培养拔尖创新型软件人才为目标，围绕软件开发与应用、软件项目管理、软件测试与质量保证等专业方向，全力探索适应经济建设与发展需求的软件人才的培养模式，为软件创新发展做出应有的贡献。

I. Professional Introduction

This major has been recruiting undergraduate students since 2005 and is a first-class engineering discipline. It has the authority to confer master's degrees in software engineering as a first-class discipline. It is also a key characteristic discipline and provincial strategic emerging (pillar) industry talent training program in Hubei Province during the 12th Five Year Plan period, a service outsourcing talent training base in Hubei Province, a collaborative education

program for "Jingchu Excellent Talents" in Hubei Province, a university level "3+1+2" top-notch talent training program, and one of the first national first-class undergraduate major construction sites. The goal of the major is to cultivate top-notch innovative software talents, focusing on professional directions such as software development and application, software project management, software testing and quality assurance, etc., fully exploring the training mode of software talents that meet the needs of economic construction and development, and making due contributions to the development of software innovation.

二、培养目标

本专业围绕学校“钢铁品质、社会英才”的人才培养总目标，培养适应国民经济发展需求，面向工业与信息化建设，德智体美劳全面发展的社会主义事业合格建设者和可靠接班人，具有良好的人文社会科学素养、社会责任感、职业道德、国际视野、创新精神，能够将软件工程领域的基础理论和专门知识用于解决复杂工程问题，能够在软件工程相关领域中，从事软件系统的研究、设计、开发、测试、维护、项目管理等工作的拔尖创新型人才，期待毕业生五年左右达到以下目标：

1. 积极投身社会主义建设事业，自觉践行社会主义核心价值观，具有良好的人文科学素养、社会责任感，遵守职业道德和规范。
2. 能够综合运用数学、自然科学和专业知识，解决在软件工程相关领域中遇到的复杂工程问题。能够设计软件工程领域复杂工程问题的解决方

案，能够在软件工程领域的复杂工程项目中胜任研究、设计、开发、测试、维护和项目管理工作。

3. 具有一定的国际视野，能在多学科、多文化背景下与业界同行及社会公众进行有效沟通和交流，能够胜任团队的核心或领导工作。

4. 能够适应国民经济发展需求、不断地自主学习，在工作中不断创新。

II. TRAINING OBJECTIVES

This major revolves around the overall goal of cultivating talents in the school's "steel quality, social talents". It aims to cultivate qualified socialist builders and reliable successors who meet the needs of national economic development, are oriented towards industrial and information construction, and have comprehensive development in morality, intelligence, physical fitness, aesthetics, and labor. They have good humanistic and social science literacy, social responsibility, professional ethics, international perspective, and innovative spirit, and can apply basic theories and specialized knowledge in the field of software engineering to solve complex engineering problems. They are also top-notch innovative talents who can engage in research, design, development, testing, maintenance, and project management of software systems in software engineering related fields. Graduates are expected to achieve the following goals in about five years:

1. Actively engage in the socialist construction cause, consciously practice the core socialist values, possess good humanistic and scientific literacy, a sense of social responsibility, and abide by professional ethics and norms.

2. Be able to comprehensively apply mathematics, natural sciences, and professional knowledge to solve complex engineering problems encountered in software engineering related fields. Capable of designing solutions to complex engineering problems in the field of software engineering, capable of conducting research, design, development, testing, maintenance, and project management in complex engineering projects in the software engineering field.

3. Having a certain international perspective, able to effectively communicate and exchange ideas with industry peers and the general public in a multidisciplinary and multicultural context, and capable of handling core or leadership tasks within a team.

4. Capable of adapting to the needs of national economic development, continuously self-learning, and constantly innovating in work.

三、专业特色（200 字左右）

本专业注重学生知识、素质和能力协调发展，坚持立德树人、理论教学与实践教学统筹兼顾的教学理念；秉承立足行业、面向社会服务的原则，建立以强基础、重应用和基础与应用并重的多层次、立体化教学体系，搭建以科技创新为示范、科研项目为引导、理论教学为基础、实践教学为依托的教学平台，在多数课程中设置了课程项目或者课程设计。在专业基础

课中培养学生的系统能力；在专业课中以实际项目为驱动，培养学生在软件生命全周期中进行需求分析、产品设计、实现、测试、维护、项目管理，从而解决复杂工程问题的能力；通过一线的理论教学、校企结合的软件开发实战、国家级大型研究项目以及互联网+、挑战杯、创青春、ACM-ICPC、数学建模等学科竞赛，全方位培养学生综合素质，培养学生的创新创业能力。近五年本专业学生在学科竞赛中获国家级、省级奖励近 100 项。本专业自 2005 年招生以来，培养几千名理论扎实、实践能力强的软件工程专门人才，本专业毕业生就业薪资和就业率连续 10 多年位于全校前列。

III. Professional characteristics

This major focuses on the coordinated development of students' knowledge, quality, and abilities, adhering to the teaching philosophy of cultivating morality and talents, and balancing theoretical and practical teaching; Adhering to the principle of being based on the industry and serving the society, we have established a multi-level and three-dimensional teaching system that emphasizes strong foundation, application, and both foundation and application. We have built a teaching platform that takes technological innovation as a demonstration, scientific research projects as guidance, theoretical teaching as the foundation, and practical teaching as the basis. In most courses, we have set up course projects or course designs. Cultivate students' systematic abilities in professional basic courses; In professional courses, driven by practical projects, cultivate students' ability to analyze requirements, design products, implement, test,

maintain, and manage projects throughout the software lifecycle, thereby solving complex engineering problems; Through front-line theoretical teaching, software development practice combining school and enterprise, national large-scale research projects, as well as Internet plus, Challenge Cup, Creative Youth, ACM-ICPC, mathematical modeling and other discipline competitions, students' comprehensive quality and innovation and entrepreneurship ability are comprehensively cultivated. In the past five years, students in this major have won nearly 100 national and provincial awards in subject competitions. Since the enrollment in 2005, this major has trained thousands of software engineering professionals with solid theoretical knowledge and strong practical abilities. The employment salary and employment rate of graduates in this major have been among the top in the school for more than 10 consecutive years.

四、毕业要求

1. 工程知识：具备数学、自然科学、计算知识，系统掌握软件工程领域的工程基础和专业知识，能够将相关知识用于解决软件工程领域的复杂工程问题。
2. 问题分析：能够综合运用数学、自然科学和工程科学的基本原理和方法，通过文献研究，对软件工程领域的复杂工程问题进行识别、表达和分析，综合考虑可持续发展的要求，以获得有效结论。
3. 设计/开发解决方案：能够综合运用理论和技术手段，针对软件工程领域的复杂工程问题提出解决方案，设计满足特定需求的软件系统、模块或开

发流程，能够在设计中体现创新性，并从健康、安全与环境、全生命周期成本与净零碳要求、法律与伦理、社会与文化等角度考虑可行性。

4. 研究：能够基于科学原理并采用科学方法对软件工程领域的复杂工程问题进行研究，制定技术路线、设计实施方案、分析解释实验数据，通过信息综合得到合理有效的结论。

5. 使用现代工具：能够针对软件工程领域的复杂工程问题，开发、选择与使用恰当的软件开发环境与工具、信息检索与分析工具，对复杂软件工程问题进行仿真与模拟，并能够理解其局限性。

6. 工程与可持续发展：能够基于软件工程相关背景知识进行合理分析，评价软件工程领域软件开发、系统设计等工程实践和复杂工程问题解决方案对健康、安全、环境、法律以及经济和社会可持续发展的影响，并理解应承担的责任。

7. 工程伦理和职业规范：有工程报国、为民造福的意识，具有良好的人文社会科学素养和较强的社会责任感，能够在软件系统设计与开发等工程实践中理解并遵守工程职业道德、规范和相关法律，履行软件工程师的社会责任。

8. 个人与团队：具有较强的团队合作意识与能力，能够正确理解在多样化、多学科背景下的团队中承担个体、团队成员以及负责人的角色，并承担其责任和义务。

9. 沟通：能够就软件工程领域的复杂工程问题与业界同行及社会公众进行有效沟通和交流，包括撰写报告和设计文稿、陈述发言、清晰表达或回应指令，并具备一定的国际视野，能够在跨文化背景下进行沟通和交流，理解、尊重语言和文化差异。

10. 项目管理：理解并掌握软件工程项目成本、进度、范围、质量、风险等管理原理与经济决策方法，并能够在多学科环境中应用。

11. 终身学习：具有自主学习、终身学习和批判性思维的意识 and 能力，能够追踪软件工程专业领域的发展动态，能够理解广泛的技术变革对工程和社会的影响，适应新技术变革，有不断学习和适应发展的能力。

IV. GRADUATION REQUIREMENTS

1. Engineering knowledge: Having knowledge in mathematics, natural sciences and computing, systematically mastering the engineering foundation and professional knowledge in the field of software engineering, and being able to apply relevant knowledge to solve complex engineering problems in the field of software engineering.

2. Problem analysis: Identify, formulate, research literature and analyse complex engineering problems in the field of software engineering reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences with holistic considerations for sustainable development.

3.Design/Development solutions: Design creative solutions for complex engineering problems in the field of software engineering and design systems, components or processes to meet identified needs with appropriate consideration for public health and safety, whole-life cost, net zero carbon as well as resource, cultural, societal, and environmental considerations.

4.Research: Conduct investigations of complex engineering problems in the field of software engineering using research methods including research-based knowledge, design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.

5.Using modern tools: Develop, select, apply and recognize limitations of appropriate software and hardware development techniques, environments and tools, information retrieval and analysis tools, including prediction and modelling, to complex engineering problems in the field of software engineering.

6.Engineering and sustainable development: When solving complex engineering problems in the field of software engineering, analyze and evaluate sustainable development impacts of the impact of engineering practices, such as software development and system design and solutions, to: society, the economy, sustainability, health and safety, legal frameworks, and the environment.

7.Professional norms: Have the consciousness of serving the country and benefiting the people through engineering, have good humanistic and social science literacy and strong social responsibility, establish and practice the core socialist values, can understand and abide by the engineering professional ethics

and norms in the design and development of computer software system and other engineering practices, and fulfill the social responsibility of software engineers.

8.Individual and team: Strong sense and ability of teamwork, able to correctly assume the roles of individual, team member and leader in diverse and inclusive teams and in multi-disciplinary, face-to-face, remote and distributed settings, and assume corresponding responsibilities and obligations.

9.Communication: Communicate effectively and inclusively with industry peers and the public on complex engineering issues in the field of software engineering, able to write reports and design documents, make presentations, clearly express or respond to instructions, and have a certain international perspective and be able to communicate and communicate in a cross-cultural context, taking into account cultural, language, and learning differences.

10.Project Management: Understand and master the management principles and economic decision-making methods of software engineering projects, such as cost, schedule, scope, quality, risk, etc., and can apply them in a multidisciplinary environment.

11.Lifelong learning: In the software engineering field, recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change.

附：培养目标实现矩阵

	培养目标 1	培养目标 2	培养目标 3	培养目标 4
毕业要求 1		√		
毕业要求 2		√		
毕业要求 3		√		
毕业要求 4		√		
毕业要求 5		√		
毕业要求 6	√			
毕业要求 7	√			
毕业要求 8			√	
毕业要求 9			√	
毕业要求 10			√	
毕业要求 11				√

五、专业主干课程

数据结构、数据库系统原理、计算机组成原理、编译原理、操作系统、计算机网络、数字逻辑与数字系统、Web 前端开发技术、Java EE 架构、移动平台软件设计、算法设计与分析、软件工程、软件设计与体系结构、软件项目管理、软件测试、团队激励与沟通。

V. Core courses

Data Structure, Principles of Database System, Principles of Computer Organization, Principles of Compiling, Operating System, Computer Network, Digital logic and Digital System, Web Front End Development Technology, Java EE Architecture, Software Design for Mobile Platform, Algorithm Design and Analysis, Software Engineering, Software Design

and Architecture, Software Project Management, Software Testing, Team Motivation and Communication.

六、基本学制：四年

VI.Recommended length of the program: 4 years

七、授予学位：工学学士

VII. Degree: Bachelor of Engineering

学生修满所规定的最低毕业学分，符合武汉科技大学授予学士学位规定，授予工学学士学位。

八、毕业学分要求：160 学分

课程类型		学分要求	课程类型		学分要求
1、公共课程平台		42	3、专业课程模块		45.5
公共基础课程		28	专业必修课程		44
通识教育课程	必修	8	专业选修课程		1.5
	选修	6	4、实践教学模块		34
2、学科基础平台		32.5	大类实践课程		3
大类学科基础 课程	必修	9	专业实践课程	必修	31
	选修	0		选修	0
专业学科基础课程		23.5	5、素质拓展模块		6

VIII、毕业学分要求：160 学分

Type of courses	Academic credits	Type of courses	Academic credits
1.CommonCourses	42	3. Specialized Courses	45.5
CommonBasicCourses	28	Required Courses	44

General Education Courses	Required Courses	8	Elective Courses		1.5
	Elective Courses	6	4. Practicum and Internship Courses		34
2. General Disciplinary Courses		32.5	Large Category Practical Courses		3
Large Category Basic Courses	Required Courses	9	Disciplinary Practical Courses	Required Courses	31
	Elective Courses	0		Elective Courses	0
Disciplinary Basic Courses		23.5	5. Quality Development Courses		6

九、 学分比例

IX、 Ratio of Credits

1. 必修选修学分比例

The proportion of compulsory elective credits

类别	学分	占总学分比例
必修	152.5	95.3125%
选修	7.5	4.6875%

2. 实践教学环节学分比例

The proportion of credits in practice teaching

类别		学分	占总学分比例
实践教学环节	实验教学学时	11.7	32.313%
	实践教学模块	34	
	素质拓展模块	6	

十、 毕业要求实现矩阵

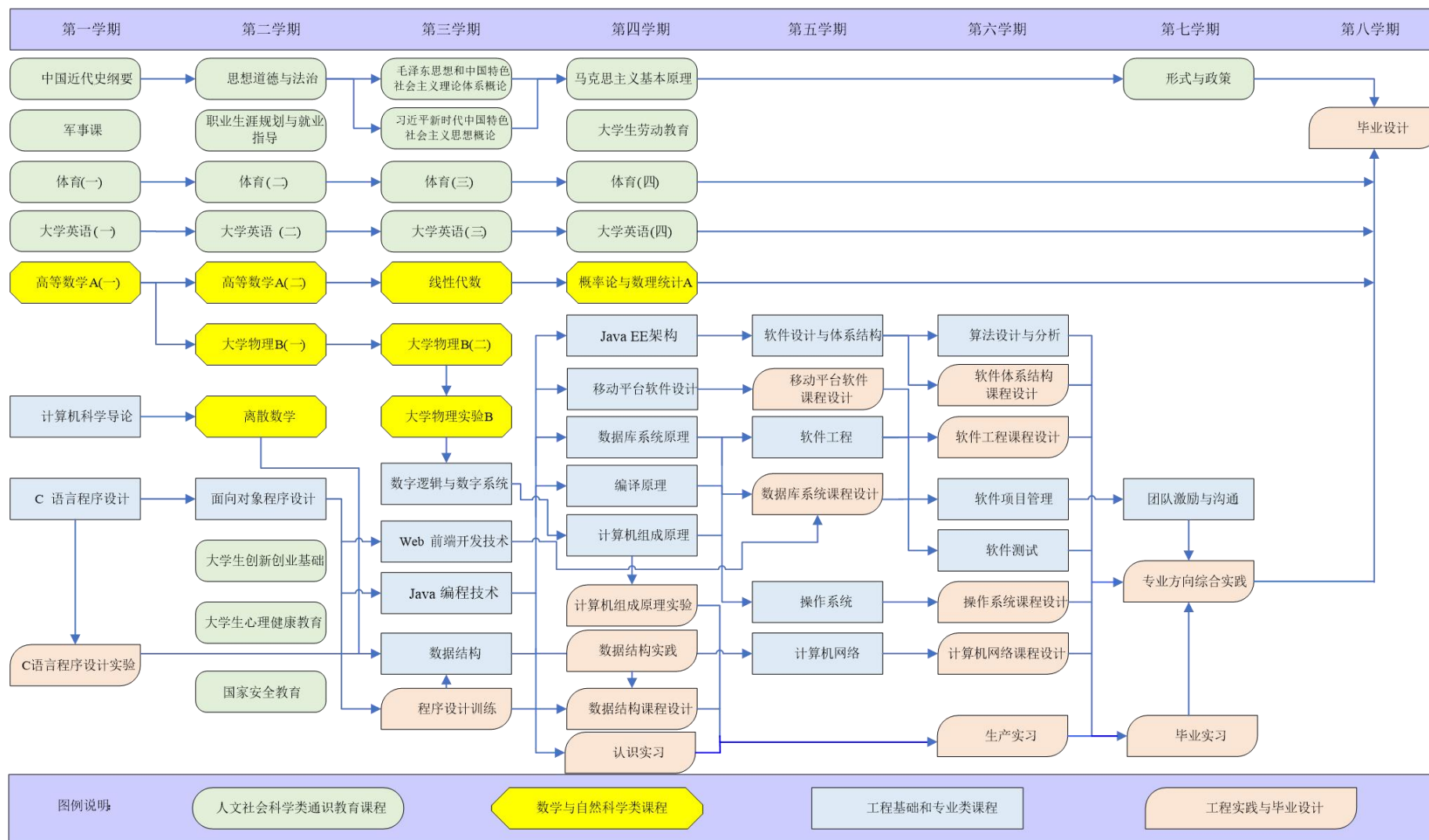
X.Graduation Realization Matrix

课程名称	软件工程专业毕业要求										
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
中国近现代史纲要							√				√
思想道德与法治			√			√	√				
国家安全教育			√			√	√				
毛泽东思想和中国特色社会主义理论体系概论							√				√
习近平新时代中国特色社会主义思想概论						√	√				√
马克思主义基本原理							√				√
形势与政策						√					√
大学英语									√		√
体育								√			
体质健康测试								√			
军事课							√	√			√
大学生心理健康教育							√				√
职业生涯规划与就业指导							√	√			√
大学生创新创业基础						√	√				
高等数学 A	√	√									
大学物理 B	√	√									
大学物理实验 B				√							
线性代数	√	√									
概率论与数理统计 A	√	√		√							
离散数学	√	√									
C 语言程序设计	√	√			√						
面向对象程序设计	√	√			√						
数据结构	√	√	√								
数字逻辑与数字系统	√	√									
计算机组成原理		√	√								

编译原理	√	√		√							
操作系统	√	√	√	√							
数据库系统原理	√	√	√								
计算机网络	√	√		√							
计算机科学导论	√	√				√					√
算法设计与分析	√			√							
Web 前端开发技术			√	√	√						
Java EE 架构			√		√						
移动平台软件设计			√	√	√						
软件设计与体系结构			√	√							
软件工程			√			√	√			√	
软件项目管理						√		√		√	
软件测试			√	√	√						
团队激励与沟通								√	√		
C 语言课程设计实验			√	√	√						
计算机组成原理实验			√	√							
程序设计训练			√	√	√						
数据结构实践			√	√	√						
数据结构课程设计			√	√					√		
数据库系统课程设计			√	√					√		
软件工程课程设计			√	√							
操作系统课程设计			√	√							
计算机网络课程设计			√			√			√		
移动平台软件课程设计			√		√						
软件设计与体系结构课程设计			√	√					√		
专业方向综合实践			√				√			√	
认识实习						√					
生产实习				√		√	√	√		√	
毕业实习						√		√		√	
毕业设计(论文)				√	√	√			√		√

创新创业实践							√	√		√	
第二课堂								√	√		
公益劳动							√	√			

十一、课程修读进程表 XI. Curriculum Progress Table



十二、教学环节设置及学分分布表

XII. Offered Course and Distribution of Academic Credits

课程类型			课程性质	课程编码	课程名称	学分	合计	课内学时			实践学时	学期	是否辅修/双学位课程	先修课程/备注
								讲课	实验	上机				
平台	公共课程平台课程	公共基础课程	必修	5103001	中国近现代史纲要 An Outline of Modern and Contemporary History of China	3	48	42	0	0	6	1		
				5105017	思想道德与法治 Ideology and morality and rule of law	3	48	42	0	0	6	2		
				5101005	毛泽东思想和中国特色社会主义理论体系概论 Theoretical System of Socialism with Chinese Characteristics	3	48	32	0	0	16	3		
				5101006	习近平新时代中国特色社会主义思想概论 An Introduction to Xi Jinping Thought on Socialism with Chinese Characteristics for a New Era	3	48	32	0	0	16	3		
				5102001	马克思主义基本原理 Fundamentals of Marxism	3	48	44	0	0	4	4		
				1401010	大学英语(一) College English (I)	2	32	32	0	0	0	1		
				1401062	大学英语(二) College English (II)	2	32	32	0	0	0	2		
				1401011	大学英语(三) College English (III)	2	32	32	0	0	0	3		
				1401064	大学英语(四) College English (IV)	1	16	16	0	0	0	4		
				1501882	体育(一) Physical Education(I)	1	26	26	0	0	0	1		
				1501883	体育(二) Physical Education(II)	1	34	34	0	0	0	2		
				1501884	体育(三) Physical Education(III)	1	34	34	0	0	0	3		
				1501885	体育(四) Physical Education(IV)	1	34	34	0	0	0	4		
				5106001	形势与政策 World Affairs and State Policy	2	64	64	0	0	0	1,2,3,4,5,6,7,8		
				1501024	体质健康测试 Physical Health Test	0	16	0	0	0	16	1-8		分散进行

	通识教育课程	必修	2502006	大学生心理健康教育 Mental Health Education	2	32	24	0	0	8	1		
			2503001	职业生涯规划与就业指导 Career Planning and Employment Guidance	1	16	16	0	0	0	2		
			2504005	军事理论 Military Theory	2	36	36	0	0	0	2		
			8001003	大学生创新创业基础 Foundation of Innovation and Entrepreneurship for college students	1	16	16	0	0	0	2		
			2501016	大学生劳动教育 College student labor education	1	32	2	0	0	30	1,2,3,4		
			5107001	国家安全教育 National Security Education	1	16	8	0	0	8	2		
		选修		理工类 Science and Technology	2								
				医学类 Medicine									
				人文社科类 Humanity and Social Science									
				经济管理类 Economic and Management									
				思想政治及新时代素质教育类 Ideological and Political Education	2								
				美育教育类 Art Education	2								
	学科基础平台课程	必修	1303004	计算机科学导论 Introduction to Computer Science	1	16	16				1		
			1303068	C 语言程序设计 C Programming	3	48	48				1	是	
			0702601	高等数学 A(一) Advanced Mathematics A(I)	5	80	80	0	0	0	1		
		必修	0702602	高等数学 A(二) Advanced Mathematics A(II)	6.5	104	104	0	0	0	2		
			0703605	大学物理 B(一) College Physics B(I)	2.5	40	40	0	0	0	2		
			0703606	大学物理 B(二) College Physics B(II)	2	32	32	0	0	0	3		
			0703607	大学物理实验 B Experiments of College Physics B	1	24	0	24	0	0	3		
			0702103	线性代数 A Linear Algebra A	3	48	48	0	0	0	3		
			0702303	概率论与数理统计 A Probability and Mathematical Statistics(A)	3	48	48	0	0	0	4		
			1303096	离散数学 Discrete Mathematics	3	48	48	0	0	0	2		
			1303098	面向对象程序设计 Object-Oriented Programming	2.5	40	32	0	8	0	2		

		软件工程 专业学科 基础课程	必修	0702602	高等数学 A(二) Advanced Mathematics A(II)	6.5	104	10 4	0	0	0	2		
				0703605	大学物理 B(一) College Physics B(I)	2.5	40	40	0	0	0	2		
				0703606	大学物理 B(二) College Physics B(II)	2	32	32	0	0	0	3		
				0703607	大学物理实验 B Experiments of College Physics B	1	24	0	24	0	0	3		
				0702103	线性代数 A Linear Algebra A	3	48	48	0	0	0	3		
				0702303	概率论与数理统计 A Probability and Mathematical Statistics(A)	3	48	48	0	0	0	4		
				1303096	离散数学 Discrete Mathematics	3	48	48	0	0	0	2		
				1303098	面向对象程序设计 Object-Oriented Programming	2.5	40	32	0	8	0	2		
		信息安全 专业学科 基础课程	必修	0702602	高等数学 A(二) Advanced Mathematics A(II)	6.5	104	10 4	0	0	0	2		
				0703605	大学物理 B(一) College Physics B(I)	2.5	40	40	0	0	0	2		
				0703606	大学物理 B(二) College Physics B(II)	2	32	32	0	0	0	3		
				0703607	大学物理实验 B Experiments of College Physics B	1	24	0	24	0	0	3		
				0702103	线性代数 A Linear Algebra A	3	48	48	0	0	0	3		
				0702303	概率论与数理统计 A Probability and Mathematical Statistics(A)	3	48	48	0	0	0	4		
				1303096	离散数学 Discrete Mathematics	3	48	48	0	0	0	2		
				1303098	面向对象程序设计 Object-Oriented Programming	2.5	40	32	0	8	0	2		
		智能科学 与技术专 业学科基 础课程	必修	0702602	高等数学 A(二) Advanced Mathematics A(II)	6.5	104	10 4	0	0	0	2		
				0703605	大学物理 B(一) College Physics B(I)	2.5	40	40	0	0	0	2		
				0703606	大学物理 B(二) College Physics B(II)	2	32	32	0	0	0	3		
				0703607	大学物理实验 B Experiments of College Physics B	1	24	0	24	0	0	3		
				0702103	线性代数 A Linear Algebra A	3	48	48	0	0	0	3		
				0702303	概率论与数理统计 A Probability and Mathematical Statistics(A)	3	48	48	0	0	0	4		
				1303096	离散数学 Discrete Mathematics	3	48	48	0	0	0	2		

				1303098	面向对象程序设计 Object-Oriented Programming	2.5	40	32	0	8	0	2		
模块	专业课程模块	计算机科学与技术专业必修课程	必修	1304065	Java 编程技术 Java Programming	2.5	40	30	0	10	0	3		
				1302133	数据结构 Data Structure	3.5	56	56	0	0	0	3	是	C 语言程序设计
				1303017	数字逻辑与数字系统 Digital Logic and Digital System	3.5	56	46	10	0	0	3	是	
				1302091	编译原理 Principles of Compiling	3	48	40	0	8	0	4		
				1302135	数据库系统原理 Principles of Database System	3	48	40	0	8	0	4	是	
				1303023	计算机组成原理 Principles of Computer Composition	3.5	56	56	0	0	0	4	是	数字逻辑与数字系统
				1307001	机器学习 Machine Learning	2	32	32	0	0	0	4		
				1302134	操作系统 Operating System	2.5	40	40	0	0	0	5	是	
				1304062	计算机网络 Computer Network	3.5	56	48	8	0	0	5	是	
				1303103	单片机技术 Single-Chip Microcomputer Technology	2.5	40	34	6	0	0	5		
				1350076	计算机视觉 Computer Vision	2.5	40	32	0	8	0	5		
				1304094	软件工程 Software Engineering	2.5	40	32	0	8	0	5		
				1303101	计算机系统结构 Computer Architecture	2.5	40	40	0	0	0	6		
				1303104	嵌入式系统 Embedded System	2.5	40	34	6	0	0	6	是	
				1303105	算法设计与分析 Algorithm Design and Analysis	2.5	40	32	0	8	0	6	是	
		软件工程专业必修课程	必修	1304065	Java 编程技术 Java Programming	2.5	40	30	0	10	0	3		
				1302133	数据结构 Data Structure	3.5	56	56	0	0	0	3		
				1303017	数字逻辑与数字系统 Digital Logic and Digital System	3.5	56	46	10	0	0	3		
				1304092	Web 前端开发技术 Web Front End Development Technology	2	32	32	0	0	0	3		
				1302091	编译原理 Principles of Compiling	3	48	40	0	8	0	4		
				1302135	数据库系统原理 Principles of Database System	3	48	40	0	8	0	4		

				1303023	计算机组成原理 Principles of Computer Composition	3.5	56	56	0	0	0	4		
				1302003	移动平台软件设计 Software Design for Mobile Platform	2.5	40	40	0	0	0	4		
				1302139	Java EE 架构 Java EE Architecture	2	32	32	0	0	0	4		
				1302134	操作系统 Operating System	2.5	40	40	0	0	0	5		
				1304062	计算机网络 Computer Network	3.5	56	48	8	0	0	5		
				1302138	软件设计与体系结构 Software Design and Architecture	2	32	32	0	0	0	5		
				1304094	软件工程 Software Engineering	2.5	40	32	0	8	0	5		
				1302004	软件项目管理 Software Project Management	2.5	40	32	0	8	0	6		
				1302136	软件测试 Software Testing	2	32	22	0	10	0	6		
				1303105	算法设计与分析 Algorithm Design and Analysis	2.5	40	32	0	8	0	6		
				1302137	团队激励与沟通 Team Motivation and Communication	1	16	16	0	0	0	7		
		信息安全专业必修课程	必修	1305038	信息安全数学基础 Mathematics Foundation of Information Security	2	32	32	0	0	0	3		
				1302133	数据结构 Data Structure	3.5	56	56	0	0	0	3		
				1303099	数字逻辑与数字系统 Digital Logic and Digital System	3	48	40	8	0	0	3		
				1302091	编译原理 Principles of Compiling	3	48	40	0	8	0	4		
				1302135	数据库系统原理 Principles of Database System	3	48	40	0	8	0	4		
				1303023	计算机组成原理 Principles of Computer Composition	3.5	56	56	0	0	0	4		
				1302134	操作系统 Operating System	2.5	40	40	0	0	0	5		
				1304065	Java 编程技术 Java Programming	2.5	40	30	0	10	0	3		
				1304062	计算机网络 Computer Network	3.5	56	48	8	0	0	5		
				1305039	密码学 Cryptography	2.5	40	40	0	0	0	4		
				1305040	软件安全 Software Security	2	32	32	0	0	0	5		

				1305044	网络安全编程与协议分析 Network Security Programming and Protocol Analysis	2	32	32	0	0	0	5		
				1303105	算法设计与分析 Algorithm Design and Analysis	2.5	40	32	0	8	0	6		
				1305041	信息系统安全 Information System Security	2	32	32	0	0	0	6		
				1305042	网络安全 Network Security	2	32	32	0	0	0	6		
				1305043	网络安全工程与实践 Network Security Engineering and Practice	2	32	32	0	0	0	6		
		智能科学与技术专业必修课程	必修	1304065	Java 编程技术 Java Programming	2.5	40	30	0	1 0	0	3		
				1302133	数据结构 Data Structure	3.5	56	56	0	0	0	3		C 语言程序设计
				1303017	数字逻辑与数字系统 Digital Logic and Digital System	3.5	56	46	10	0	0	3		
				1304095	人工智能 Artificial Intelligence	2	32	32	0	0	0	3		
				1302091	编译原理 Principles of Compiling	3	48	40	0	8	0	4		
				1302135	数据库系统原理 Principles of Database System	3	48	40	0	8	0	4		
				1303023	计算机组成原理 Principles of Computer Composition	3.5	56	56	0	0	0	4		数字逻辑与数字系统
				1307001	机器学习 Machine Learning	2	32	32	0	0	0	4		
				1302134	操作系统 Operating System	2.5	40	40	0	0	0	5		
				1304062	计算机网络 Computer Network	3.5	56	48	8	0	0	5		
				1350076	计算机视觉 Computer Vision	2.5	40	32	0	8	0	5		机器学习
				1304094	软件工程 Software Engineering	2.5	40	32	0	8	0	5		
				1307002	自然语言处理 Nature Language Processing	2.5	40	40				6		机器学习
				1307004	机器人理论与技术 Robotic Theory and Technology	2.5	40	40				6		
				1303105	算法设计与分析 Algorithm Design and Analysis	2.5	40	32	0	8	0	6		
		专业选修	选修	1304046	网络科学与工程导论 Introduction to Network Science and Engineering	2	32	32	0	0	0	2		

课程	1305030	网络空间安全导论 Introduction to Cybersecurity	2	32	32	0	0	0	2		
	1304092	Web 前端开发技术 Web Front End Development Technology	2	32	32	0	0	0	3		
	1304095	人工智能 Artificial Intelligence	2	32	32	0	0	0	3		推荐 软件 工程 专业
	1307001	机器学习 Machine Learning	2	32	32	0	0	0	4		
	0702136	数学建模 Mathematical Modeling	2	32	24	0	8	0	5		
	1302144	数据库应用技术 Database Application Technology	2	32	28	0	4	0	5		
	1303106	移动应用开发 Development of Mobile Application	2	32	24	8	0	0	5		
	1303107	数值计算基础 Basics of Numerical Calculation	2	32	32	0	0	0	5		推荐 智能 科学 与技 术专 业
	1303108	图像处理 Image Processing	2	32	24	0	8	0	5		
	1304090	知识图谱技术与应用 Knowledge graph technique and application	2	32	32	0	0	0	5		
	1305040	软件安全 Software Security	2	32	32	0	0	0	5		
	1302139	Java EE 架构 Java EE Architecture	2	32	32	0	0	0	6		
	1302140	数据挖掘 Data Mining	2	32	32	0	0	0	6		推荐 软件 工程 专业
	1302145	Linux 内核与程序设计 Linux Kernel and Programming	2	32	32	0	0	0	6		推荐 软件 工 程、 智能 科学 与技 术专 业
	1303095	虚拟现实与仿真 Virtual Reality and Simulation	2	32	24	8	0	0	6		
	1303102	接口与通讯技术 Interface and Communication Technology	2.5	40	34	6	0	0	6		

实践教学模块	大类实践课程	必修	1303109	专业英语 Subject-based English	2	32	32	0	0	0	6		
			1304070	大数据技术原理与应用 Big Data Technology and Applications	2	32	26	0	6	0	6		
			1305017	信息内容安全 Information content Security	2	32	24	0	8	0	6		
			1304093	动手学深度学习 Hands-on Deep Learning	2	32	32	0	0	0	3		推荐 软件 工程 专业
			1305020	信息隐藏 Information Hiding	2	32	24	0	8	0	5		
			1305033	信息论与编码 Information Theory and Coding	2	32	32	0	0	0	5		
			1305045	可编程网络技术与安全 Programmable Network Technology and Security	2	32	32	0	0	0	5		
	计算机科学与技术专业实践课程	必修	1303008	C 语言程序设计实验 Experiments in C Programming	1	24	0	0	2 4	0	1		
			2504006	军事训练 Military Training	2	112	0	0	0	112	1		
		必修	1303015	程序设计训练 Advanced Training for Programming Design	1	1(周)	0	1(周)	0	0	3		
			1302006	数据结构实践 Data Structure Practice	1	1(周)	0	1(周)	0	0	4		
			1302034	数据结构课程设计 Course Project in Data Structure	1	1(周)	0	0	0	1(周)	4		
			1303009	计算机组成原理实验 Experiments in Principles of Computer Organization	1	24	0	24	0	0	4		
			1305025	认识实习 Cognition Practice	1	1(周)	0	0	0	1(周)	4	是	
			1302132	数据库系统课程设计 Course Project in Database System	1	1(周)	0	0	0	1(周)	5		
			1302122	操作系统课程设计 Course Project in Operating System	1	1(周)	0	0	0	1(周)	6		
			1303036	软件开发技能实习 Practice Experience in Software Development Skills	2	2(周)	0	0	0	2(周)	5		
			1302131	软件工程课程设计 Course Project in Software Engineering	1	1(周)	0	0	0	1(周)	6		
			1303067	单片机技术课程设计 Course Project in Single-Chip Microcomputer Technology	1	1(周)	0	0	0	1(周)	6		
			1304022	计算机网络课程设计 Course Project in Computer Network	1	1(周)	0	0	0	1(周)	6		

				1303006	计算机系统综合实践 Comprehensive Practice in Computer System	1	1(周)	0	0	0	1(周)	7		
				1303065	嵌入式系统课程设计 Course Project in Embedded System	1	1(周)	0	0	0	1(周)	7		
				1303097	生产实习 Production Practice	2	2(周)	0	0	0	2(周)	6	是	
				1303055	毕业实习 Pre-graduation Internship	2	2(周)	0	0	0	2(周)	7		
				1303057	毕业设计(论文) Undergraduate Project(Thesis)	14	14(周)	0	0	0	14(周)	8	是	
		软件工程 专业 实践 课程	必修	1303015	程序设计训练 Advanced Training for Programming Design	1	1(周)	0	1(周)	0	0	3		
				1302006	数据结构实践 Data Structure Practice	1	1(周)	0	1(周)	0	0	4		
				1302034	数据结构课程设计 Course Project in Data Structure	1	1(周)	0	0	0	1(周)	4		
				1303009	计算机组成原理实验 Experiments in Principles of Computer Organization	1	24	0	24	0	0	4		
				1305025	认识实习 Cognition Practice	1	1(周)	0	0	0	1(周)	4		
				1302014	移动平台软件课程设计 Software Design for Mobile Platform	1	1(周)	0	0	0	1(周)	5		
				1302132	数据库系统课程设计 Course Project in Database System	1	1(周)	0	0	0	1(周)	5		
				1302122	操作系统课程设计 Course Project in Operating System	1	1(周)	0	0	0	1(周)	6		
				1302124	软件体系结构课程设计 Course Project in Software Architecture	1	1(周)	0	0	0	1(周)	6		
				1302131	软件工程课程设计 Course Project in Software Engineering	1	1(周)	0	0	0	1(周)	6		
				1304022	计算机网络课程设计 Course Project in Computer Network	1	1(周)	0	0	0	1(周)	6		
				1302026	专业方向综合实践 Major Comprehensive Practice	2	2(周)	0	0	0	2(周)	7		
				1303097	生产实习 Production Practice	2	2(周)	0	0	0	2(周)	6		
				1303057	毕业设计(论文) Undergraduate Project(Thesis)	14	14(周)	0	0	0	14(周)	8		
				1305027	毕业实习 Pre-graduation Practice	2	2(周)	0	0	0	2(周)	7		
	信息	必		1305025	认识实习 Cognition Practice	1	1(周)	0	0	0	1(周)	1		

		安全 专业 实践 课程	修	1303015	程序设计训练 Advanced Training for Programming Design	1	1(周)	0	1(周)	0	0	2		
				1302006	数据结构实践 Data Structure Practice	1	1(周)	0	1(周)	0	0	3		
				1302034	数据结构课程设计 Course Project in Data Structure	1	1(周)	0	0	0	1(周)	4		
				1302132	数据库系统课程设计 Course Project in Database System	1	1(周)	0	0	0	1(周)	4		
				1303009	计算机组成原理实验 Experiments in Principles of Computer Organization	1	24	0	24	0	0	4		
				1302122	操作系统课程设计 Course Project in Operating System	1	1(周)	0	0	0	1(周)	5		
				1305006	密码学课程设计 Course Project in cryptology	1	1(周)	0	0	0	1(周)	5		
				1303097	生产实习 Production Practice	2	2(周)	0	0	0	2(周)	6		
				1304022	计算机网络课程设计 Course Project in Computer Network	1	1(周)	0	0	0	1(周)	6		
				1305037	软件安全课程设计 Course Project in Software Security	1	1(周)	0	0	0	1(周)	6		
				1305022	网络安全课程设计 Course Project in Network Security	1	1(周)	0	0	0	1(周)	7		
				1305027	毕业实习 Pre-graduation Practice	2	2(周)	0	0	0	2(周)	7		
				1305031	信息安全综合实践 Comprehensive Practice Experience of Information Security	1	1(周)	0	0	0	1(周)	7		
				1303057	毕业设计(论文) Undergraduate Project(Thesis)	14	14(周)	0	0	0	14(周)	8		
		智能 科学 与技 术专 业	必修	1307005	自然语言处理实验 Experiments in Nature Language Processing	1	24		24		24	6		
				1307006	机器人理论与技术实验 Experiments in robotic theory and technology	1	24		24		24	6		
				1305025	认识实习 Cognition Practice	1	1(周)	0	0	0	1(周)	1		
				1303015	程序设计训练 Advanced Training for Programming Design	1	1(周)	0	1(周)	0	0	2		
				1302006	数据结构实践 Data Structure Practice	1	1(周)	0	1(周)	0	0	3		

			1302034	数据结构课程设计 Course Project in Data Structure	1	1(周)	0	0	0	1(周)	3		
			1302132	数据库系统课程设计 Course Project in Database System	1	1(周)	0	0	0	1(周)	4		
			1303009	计算机组成原理实验 Experiments in Principles of Computer Organization	1	24	0	24	0	0	4		
			1302122	操作系统课程设计 Course Project in Operating System	1	1(周)	0	0	0	1(周)	5		
			1302131	软件工程课程设计 Course Project in Software Engineering	1	1(周)	0	0	0	1(周)	6		
			1304022	计算机网络课程设计 Course Project in Computer Network	1	1(周)	0	0	0	1(周)	5		
			1303097	生产实习 Production Practice	2	2(周)	0	0	0	2(周)	6		
			1303055	毕业实习 Pre-graduation Internship	2	2(周)	0	0	0	2(周)	7		
			1303057	毕业设计(论文) Undergraduate Project(Thesis)	14	14(周)	0	0	0	14(周)	8		
素质拓展模块	创新创业教育	必修		创新创业实践 3 学分 Innovation Practices 3 Academic Credits									
	第二课堂			第二课堂 3 学分 Second Classroom 3 Academic Credits									

十三、教学进程安排表

XIII. Teaching Progress Table

学期	周 次																											
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28
1	♀	♀	☉★	★	★	□	□	□	□	□	□	□	□	□	□	□	□	□	●									
2	□	□	□	□	□	□	□	□	□	□	□	□	□	□	□	□	□	□	●									
3	□	□	□	□	□	□	□	□	□	□	□	□	□	□	□	□	□	□	●									
4	≡	×	□	□	□	□	□	□	□	□	□	□	□	□	□	□	□	□	●									
5	×	×	□	□	□	□	□	□	□	□	□	□	□	□	□	□	□	□	●									
6	/	/	×	×	□	□	□	□	□	□	□	□	□	□	□	□	□	□	●									
7	#	#	□	□	□	□	□	□	□	□	□	□	□	□	□	□	□	×	●									
8	※	※	※	※	※	※	※	※	※	※	※	※	※	※	※	※	※	+	√	√								

符号说明:

- 1、♀ 入学前机动 2、☉ 入学教育 3、★ 军训 4、□ 理论教学 5、√ 机动时间 6、● 考试 7、× 课程设计 8、E 专业实验或实习 9、— 假期
 10、▲ 学年论文 11、G 技能训练 12、※ 毕业设计(论文) 13、+ 毕业鉴定 14、# 毕业实习 15、S 写生 16、/ 生产实习(金工实习)
 17、T 教材教法 18、☆ 教育实习 19、○ 技能教育实习 20、◎ 专题讲座 21、◆ 公益劳动 22、△ 社会调查 23、≡ 认识实习